

What's in the box? Exploring the (non-)predictability of ChatGPT's scoring of student writing

Abstract

Generative artificial intelligence (GenAI) tools are increasingly marketed to faculty for grading and feedback, yet little is known about the criteria these systems use when assigning scores to student writing. In alignment with writing analytics' emphasis on construct validity and transparent scoring, this study investigates the grading tendencies of ChatGPT 3.5 by examining whether linguistic features traditionally predictive in automated essay scoring (AES)—including lexical sophistication, syntactic complexity, and cohesion—account for variance in AI-generated scores. Using data from a larger project on AI-enhanced writing feedback across nine courses in multiple disciplines at a STEM-focused university, we applied regression models (linear, lasso, and ridge) with linguistic indices drawn from TAALES, TAASSC, TAACO, and TAALED. Results indicate that ChatGPT's grading is not random, as Kolmogorov-Smirnov tests rejected alignment with skewed distributions; however, regression models failed to identify reliable linguistic predictors of AI scores, with R^2 values below zero across all approaches. These findings suggest that ChatGPT's scoring practices cannot be reliably predicted using the same linguistic features that have historically predicted AES and human scoring practices. This divergence highlights a core writing-analytics concern: whether automated systems embody interpretable constructs that support fair, pedagogically meaningful evaluation. We argue that understanding these differences is important as educators and institutions consider the role of GenAI systems in writing assessment in the future.

Keywords: Automated Grading; Linguistic Features; Generative AI; Large Language Models

1. Introduction

The impacts of generative AI (GenAI) have been felt across all aspects of higher education in the past three years, with increasing attention being paid to the diverse ways GenAI might be integrated into instruction across the curriculum (Baruah & Baruah, 2024; Dwivedi et al., 2023; Mavundla et al., 2025). In particular, the use of GenAI, including virtual assistants, for automated feedback and essay grading has become a focal point (Bulut et al., 2024; Guo & Wang, 2024; Kaliisa et al., 2025; Rudniy, 2024). Whether through general-purpose GenAI interfaces like Microsoft Copilot, Google Gemini, or ChatGPT, or more specialized educational technologies like EssayGrader, Markr, or LMS-integrated tools, an increasing number of faculty members are experimenting with uses of GenAI for feedback and evaluation (Bent et al., 2025; Heinrich et al., 2025).

Even as the availability and usage of GenAI tools by faculty are increasing, much remains unknown about the precise nature of the feedback or assessment GenAI tools provide. Many current studies focus on the effectiveness of GenAI grading as defined in terms of students' and teachers' perception or uptake of AI feedback (Hall et al., 2024; Zhai & Ma, 2023), while others have compared GenAI feedback and scores against those provided by human raters (Almegren et al., 2024; Mizumoto & Eguchi, 2023). Yet, for the most part, the processes by which GenAI tools evaluate and provide feedback to writers are hidden inside a "black box," and researchers have noted that the lack of transparency and explainability in automated grading

systems can negatively impact their usability (Hall et al., 2024). While prior research has indicated that GenAI-assigned scores for writing are not random and that feedback outputs can be shaped by reinforcement training provided by instructors (Velez et al., 2025), we argue for the need to better understand the criteria GenAI tools use to evaluate written texts.

These concerns map directly onto ongoing priorities in writing analytics: scholars argue that automated tools must be positioned in relation to construct validity and ethical practice rather than accepted as neutral “efficiency” solutions. Writing analytics leaders have urged that analytics work be construct-based and mindful of ethical harms (e.g., bias, privacy, misuse), not simply be a technical substitute for human judgment (Lang et al., 2019; Palmquist, 2019). The WA taxonomy identified *Ethical Philosophy* as a core practice domain, highlighting “concerns about bias in analytics, misuse of data, privacy concerns, and ownership of student data/student rights” as central challenges for the field (Lang et al., 2019). Similarly, Palmquist (2019) cautioned that predictive tools used without attention to their “limitations and built-in biases” carry “significant risk of biasing our understanding of individual students and groups of students” (p. 8). Furthermore, as Norbert Elliott shared (as cited by Palmquist, 2019, p. 5), “Learning analytics is not construct-based. Writing analytics is.” This distinction raises important questions about the validity and fairness of LLM scoring, which are questions this study takes up by examining the extent to which LLM-assigned scores relate to interpretable linguistic constructs grounded in writing research.

To investigate this question of construct alignment, this study examines whether linguistic features traditionally predictive in automated essay scoring (AES)—including lexical sophistication, syntactic complexity, and cohesion—account for variance in AI-generated scores. Drawing on data from a 2023–2024 pilot study of AI-enhanced writing feedback in courses across disciplines, the authors applied regression models (linear, lasso, and ridge) with linguistic indices drawn from the Tool for the Automatic Analysis of Lexical Sophistication (TAALES), the Tool for the Automatic Analysis of Syntactic Sophistication and Complexity (TAASSC), the Tool for the Automatic Analysis of Cohesion (TAACO), and the Tool for the Automatic Analysis of Lexical Diversity (TAALED) to identify potential patterns in the criteria used to evaluate and respond to student writing samples in this set.

2. Literature Review

2.1 Automated Essay Scoring

Automated essay scoring (AES) and automated writing evaluation (AWE) systems have long been a focal point in educational assessment research due to their promise in providing scalable, consistent, and rapid evaluation of students’ writing. Early systems such as e-rater by the Educational Testing Service (Attali & Burstein, 2006) and IntelliMetric Essay Scoring System by Vantage Learning (Rudner et al., 2006) relied on statistical modeling of linguistic features (e.g., word counts and prompt-specific vocabulary used) to approximate the judgments of trained human raters. Linguistic features used in feature-based AES systems typically fall into three major categories: lexical, syntactic, and cohesion-related (e.g., Crossley et al., 2016; Kyle et al., 2018; Kyle & Crossley, 2015). Measures of lexical sophistication typically capture features such as word frequency, academic vocabulary use, and distributions of content versus function words. TAALES captures over 400 indices of lexical sophistication (Kyle et al., 2018; Kyle & Crossley, 2015) and has demonstrated strong predictive power, with Kyle et al. (2018) reporting that

lexical indices from TAALES 2.0 accounted for up to 58% of the variance in lexical proficiency scores for narrative essays. Similarly, TAASSC measures indices related to syntactic development, including measures of clausal and phrasal elaboration, such as *Verb Phrases per T-unit* and *Dependents per Nominal*, for example (Kyle, 2016). For cohesion and coherence, TAACO calculates 150 indices related to word overlap, semantic similarity, and connective use at the local, global, and overall levels (Crossley et al., 2016). Crossley et al. (2016) also found that global cohesion features, such as paragraph-level noun overlap, positively predicted expert ratings of coherence, and Crossley and Kim (2022) similarly found paragraph-level noun and pronoun overlap to be a high predictor of writing quality.

These systems offered an efficient solution for large-scale assessments such as the Graduate Management Admission Test (GMAT) and the Test of English as a Foreign Language (TOEFL) by enabling faster turnaround times for scores and more consistent application of scoring rubrics, with even higher scoring agreement than two independent human raters in some cases (Attali & Burstein, 2006). Yet, these systems were often critiqued by writing studies scholars who raised concerns about the validity of the scoring mechanisms and the apparent opacity of the scoring processes (Haswell, 2006; Herrington & Moran, 2009; James, 2007; McCurry, 2010; Yancey et al., 2004). Thus, automated scoring and feedback systems were not widely adopted within writing studies for small-scale classroom assessment.

More recently, the rise of artificial intelligence and large language models (LLMs) has substantially expanded the capabilities of AES systems, and the relative ease of access to these tools has increased the potential (and in some cases, pressure) for individual instructors to incorporate computer-generated feedback and scoring into their courses. As Bulut et al. (2024) noted, AI-enabled scoring technologies can now facilitate the automated evaluation of writing tasks with high consistency and immediacy, while offering rich, detailed feedback and reducing educator workload. Rudniy (2024) also argued that integration of automated scoring and feedback can produce an “initial phase” score to facilitate more efficient and potentially fairer human scoring processes. While traditional AES systems clearly relied on engineered linguistic features to drive scoring decisions, it remains an open question whether newer AI-based systems—particularly those built on LLMs like ChatGPT—operate on similar underlying principles. This study investigates that very issue: whether quantifiable linguistic features hold predictive value for AI-generated scores.

2.2 Grading Performance of AES and GenAI Tools

Traditional AES models have yielded fairly high predictive accuracy using the types of linguistic features described above. Vajjala (2018) demonstrated that the relative importance of different features can vary significantly depending on the writing prompt, dataset, and learner population. For example, in one dataset, certain lexical sophistication measures (like usage of advanced vocabulary) might have been top predictors of a high score, whereas in the other dataset, error rates or syntactic complexity measures might have played a larger role. This indicates that the importance of specific linguistic features is context-dependent. The results suggested that there was no single universal subset of features that consistently dominated AES performance; instead, the writing prompt context, scoring guidelines, and learner populations could shift which aspects of language were most salient for the score. Despite these differences, overall, the study’s feature-based models achieved strong correlations with human scores.

Building from this previous line of work, researchers have recently begun to examine how LLMs like ChatGPT perform on grading tasks, often by studying their performance in

comparison to human raters and to traditional AES systems. Awidi (2024) investigated the effectiveness of ChatGPT in grading short reflective essays, comparing its performance to that of human tutors. The mixed-methods study found a high level of agreement between ChatGPT and the expert tutors in overall grades assigned, with ChatGPT showing strong inter-rater reliability (Cohen's kappa = 0.83). Mizumoto and Eguchi (2023) explored GPT-3's performance on a corpus of second-language learner essays. Their findings showed that GPT-3 alone achieved fair to moderate agreement with human scores (Quadratic Weighted Kappa = .388). However, when combined with linguistic features from tools such as TAALES and TAACO, the predictive accuracy of the regression models improved significantly (Quadratic Weighted Kappa = .605).

Recent scholarship has also compared the accuracy of different AI tools for automated scoring and feedback. Latif and Zhai (2024) studied the efficacy of automated scoring across two different systems, Google's BERT model and a fine-tuned ChatGPT, using a diverse set of secondary-level writing samples representing six assessment tasks. Their results indicated a significantly higher scoring accuracy for the fine-tuned GPT-3.5 vs. BERT, with an average accuracy increase of 9.1% across all tasks. Further research by Almegren et al. (2024) also examined different AI tools' ability to score and provide feedback on English-as-a-foreign-language writing. The model generated detailed, rubric-aligned comments across dimensions such as content, vocabulary, organization, and grammar. However, they found that ChatGPT, Gemini, and Bing consistently assigned lower scores than human raters, which pointed to a potential systematic undervaluation of certain discourse patterns. Despite consistent internal logic across categories, its evaluations did not always align with those of instructors. Interestingly, the study noted that the AI tool's scores in specific categories did not diverge much from its own overall assessment of the essay. In other words, if the AI tool gave a low overall score, it generally gave proportionally low scores in each category, whereas human raters might sometimes differentiate (e.g., strong content but weaker grammar). Additionally, the AI provided more elaborated feedback comments than the humans typically did, presumably because the AI can generate lengthy explanations or suggestions, whereas human feedback was shorter, a finding echoed by Guo and Wang (2024) in their study comparing GenAI feedback to that of EFL lecturers. These results again point to the existence of some pattern in how GenAI scores student work, but they do not identify what this pattern is.

2.3 The Present Study

Despite growing interest in ChatGPT's applications in education, few studies thus far have systematically examined the linguistic features that may drive its grading behavior. Mizumoto and Eguchi (2023) made strides in this area by combining GPT-generated scores with linguistic indices to predict human-assigned proficiency levels in second-language writing. While our study adopts a similar methodological approach by using regression models to link linguistic features with AI-generated scores, it differs in key respects. Rather than evaluating how well ChatGPT predicts human ratings, we focus on identifying whether any linguistic features that commonly predict human scoring can explain the scores that ChatGPT itself assigns.

Understanding how the scores are assigned is necessary to know how these tools should or could be applied effectively. Additionally, our dataset spans multiple academic disciplines and student proficiency levels, which allows us to explore grading tendencies in a more varied context. By using linguistic indices from TAALES, TAASSC, TAALED, and TAACO, we aim to explore whether there is a linguistic basis that can explain ChatGPT's scoring decisions across academic contexts. With these goals in mind, the study was guided by the following research questions:

1. Is there evidence that ChatGPT 3.5/4o assigns random grades to student writing?
2. Can a model using *a large set of linguistic features* traditionally used in Automated Essay Scoring be used to predict scores that ChatGPT 3.5 assigns to student writing?
3. Can a model using *a select subset of linguistic features* traditionally used in Automated Essay Scoring be used to predict scores that ChatGPT 3.5 assigns to student writing?
4. Can a model using *a linear combination of linguistic features* traditionally used in Automated Essay Scoring be used to predict scores that ChatGPT 3.5 assigns to student writing?
5. Can a classification model using linguistic features traditionally used in Automated Essay Scoring be used to predict letter grades that ChatGPT 3.5 assigns to student writing?

Our hope is that this analysis will contribute to broader conversations about AI reliability, the interpretability of LLM output, and the extent to which ChatGPT's assessments replicate or diverge from established AES systems.

3. Methodology

We explicate the methods employed for this study in two primary sections. First, we discuss the methods used to generate the dataset on which we based our statistical analyses for this article. We have previously reported on preliminary qualitative analyses of this same data set (Anonymized, 2025). As such, more information about the methods described in this first section can also be found in Anonymized (2025). We then explain the statistical analyses used to generate the new quantitative findings on which we report in this article.

The results reported in this paper are taken from a larger project with two major foci: 1) analyzing students' perspectives on the use of LLMs for automated writing assignment scoring and feedback, and 2) identifying commonalities and differences among instances of AI-automated feedback on writing assignments across various academic disciplines. A primary goal of the project, in line with the first area of focus, was exploring students' perspectives on the use of automated feedback and grading by instructors in classrooms. This goal motivated various methodological decisions, including the ways in which instances of automated feedback and grading were obtained via instructor use of LLMs. Because the primary goal of the current article is to examine the alignment of the GenAI scoring with known measures of human raters, we will primarily describe the ways in which we generated and analyzed the AI scoring. As such, other than briefly describing the steps taken to protect student data privacy below, we will then refrain from referencing any methodological processes pertaining to the collection and analysis of student-facing data.

3.1 Data Collection – Courses and Assignments

Members of the research team generated instances of automated feedback and grading from courses they taught in a variety of disciplines and course levels. In total, data was collected from nine courses, undergraduate and graduate levels, at a large private, STEM-focused university. These courses spanned the following disciplines: 1) Emergency Services, 2) Computer Science, 3) Mathematics, 4) Homeland Security and Human Resilience, 5) Humanities, and 6) Communication.

In each course from which data was collected, the project member chose a single, low-stakes written assignment for which to obtain automated feedback and evaluation from an LLM that would be shared with students but not used for course grading (students were also given the option to not have their assignment reviewed by AI). The project member then collected and anonymized all submissions from the assignment in text form for processing. In line with the broad goal of reflecting the kind of non-specialized AI feedback and evaluation in which many classroom instructors might choose to engage, the project team used the platform ChatGPT (OpenAI, 2023) to process the data. At the time of the initial data collection, ChatGPT 3.5 was the default version, and only one project member had access to ChatGPT 4o.

Project members in all disciplines entered the text of the student submissions into ChatGPT after providing the rubric and assignment information. Again, in pursuit of non-specialized instructor simulation, there was not a pre-determined unified method for use of ChatGPT other than provision of rubric, assignment, and submission information. There was some variety in the process for use of ChatGPT, such as the level of detail in the instructor-created prompts and the frequency with which ChatGPT was re-prompted with the rubric and assignment information. The team member prompted ChatGPT to both provide feedback on the assignment and assign a score to it. The generated instances often included scores both on individual rubric elements as well as total assignment scores. Only the total assignment scores were used for the statistical tests described below.

The choice of ChatGPT was strategic for multiple reasons, including its simulation of open-access use by instructors and its ubiquity in the AI landscape. Moreover, ChatGPT is a generalist model, as opposed to other systems such as EssayGrader or Markr that might employ multiple more targeted machine learning algorithms to accomplish different tasks such as grading or feedback. Similar to Awidi (2024), our generalist approach also allowed the project team to engage in more task-specific training and control over the ways in which the model is prompted and utilized.

3.2 Analysis Methods

We employed several statistical tests and models to investigate potential patterns in ChatGPT's assignment of student grades. All analysis was conducted in Python and is available at the published GitHub repository: [https://www.github.com/\[anonymized for peer review\]](https://www.github.com/[anonymized for peer review]).

The one-sample Kolmogorov-Smirnov (K-S) tests whether a sample resulted from a given probability distribution (Daniel, 1990). The test measures the difference in the empirical distribution function of the sample and the cumulative distribution function of the reference probability distribution. The null hypothesis is that the sample does come from the reference probability distribution and thus a low p-value provides evidence that the sample does not come from the reference probability distribution.

Linear regression models were developed under three conditions: no regularization, L1 regularization (lasso), and L2 regularization (ridge). Regularization is used to prevent overfitting of models to sample data when dealing with a large number of predictive values (Hastie et al., 2013). Both lasso and ridge regularization shrink the regression coefficients by imposing a penalty term. The lasso penalty is proportional to the absolute value of the coefficients while the ridge penalty is proportional to the square of the coefficients. Lasso regularization results in a model where most predictors are set to 0 (providing a form of automatic feature selection) while ridge regularization provides non-zero weights to each predictor. As we do not make

assumptions on the number of import features for the regression models, it is appropriate to apply both regularizations and compare the results.

Regression models were evaluated based on their R-squared (R^2) score. R^2 is the coefficient of determination and describes the proportion of variation that is predictable from the features. Unlike a coefficient of determination obtained as the square of a simple correlation coefficient, an R^2 score for linear regressions can be negative. This occurs as the predictive model is compared against using the mean of the model as a sole predictor and is illustrative of a particularly poor model (James et al., 2023).

Model validation occurred using five-fold cross-validation. This method starts by randomly segmenting the data into five equal groups. For each group, it constructs a model using four of these groups, applies the model to the fifth group, and calculates the R^2 score. After calculating the R^2 score for each model, it then provides the average score across all five models. This ensures all data is used to validate models and reduces the variance that can occur when using only one holdout set of data to validate a model.

The SelectKBest method from SciKit-Learn determines the best k features for regressions and classifications based on the chosen parameter (Pedregosa et al., 2011). We used the `f_regression` parameter to rank all features by first calculating the cross correlation between each regressor and the target regressor, then converting this to an F score for ranking features. Principal Component Analysis (PCA) was used for feature projection. PCA creates n linear combinations of original features to reduce the number of features used on the model while attempting to maintain as much data from the original set of features as possible (James et al., 2023). Regression models were developed using PCA sets of two to eight linear combinations of features.

Random Forest models were developed using the GridSearchCV method from SciKit-Learn. The grid search considered the “`max_depth`” and “`n_estimator`” parameters when searching for an optimal random forest on the selected data based on accuracy scores. Max depth options of 2, 4, 6, and 8 were used to help alleviate issues with random forests overfitting on the training data. N estimators of 10, 20, 30, 40, and 50 were used to allow for varying numbers of decision trees to be used in a single random forest to also prevent overfitting on the training data. All random forests used class weights to be inversely balanced by the number of targets in the predicted classes.

4. Results

Overall, it remains unclear how ChatGPT 3.5/4o assigned student grades. K-S hypothesis tests suggested strong evidence that assigned student grades did not come from a skewed distribution similar to the sample, and thus, we have evidence to suggest ChatGPT does not assign grades randomly. All variations of linear regression models failed to adequately predict assigned student grades, with nearly all models performing worse than a constant mean linear model. Even when relaxing the predictive target to categorical letter grades, no acceptable predictive model emerged. This is strong evidence that ChatGPT’s grading patterns cannot be reliably predicted using linguistic features that commonly predict instructor grades on student assignments. The following sections provide details on specific results related to each research question. Complete results and corresponding scripts to run analysis can be found in the GitHub repository: [link removed; anonymized for peer review].

4.1 Test of Sample from Random Distribution

To search for a predictive model to describe how ChatGPT 3.5 assigns students' grades, we first investigated whether the assigned scores were from a random distribution. A histogram of the distribution of student grades, as seen in Figure 1, suggests a negatively-skewed random distribution. We thus performed a Kolmogorov-Smirnov test for goodness of fit for the assigned grades versus random negatively skewed distributions. As we do not know the mean, standard deviation, and skew of all potential ChatGPT-assigned grades, we performed an exhaustive construction of negatively skewed distributions for the following: (a) means between 0.75 and 0.95, (b) standard deviations between 0.05 and 0.25, and (c) skews between -1 and -2. Mean and standard deviation were iterated by 0.01 while skew was iterated by 0.02, resulting in 20,000 different negatively-skewed distributions. Ranges for the mean, standard deviation, and skew were chosen based on their sample values of 0.87, 0.10, and -1.48, respectively.

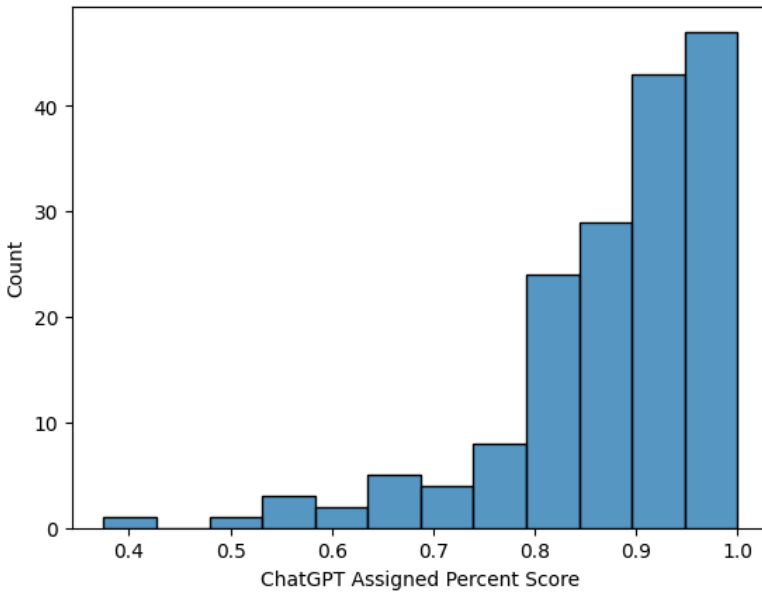


Figure 1: Histogram of the distribution of ChatGPT-assigned student grades on discussions with 0.87 mean, 0.10 standard deviation, and -1.48 skew.

All 20,000 K-S hypothesis tests rejected the null hypothesis that the ChatGPT assigned grades were from the specific negatively-skewed distribution with p-values less than $1E-10$. This is strong evidence that the ChatGPT-assigned grades were not randomly assigned from a skewed distribution and that some pattern may exist to predict how ChatGPT assigned grades.

4.2 Regression Models on All Features

To investigate whether linguistic features could be used to predict how ChatGPT assigned grades, three types of linear regressions were run on each subset of linguistic features extracted from TAACO (42), TAALED (18), TAALES (241), and TAASSC (355). Average R^2 score from the five-fold cross-validation for each linear regression is presented in Table 1.

Table 1

Average R^2 score from a five-fold cross-validation for each linear regression built from the four subsets of linguistic features.

	TAACO	TAALED	TAALES	TAASSC
# of features	42	18	241	355
Linear (no reg)	-0.133	-0.197	-6.274	-3.377
Lasso (L1 reg)	-0.015	-0.019	0.201	-0.033
Ridge (L2 reg)	0.113	-0.08	-0.563	-0.766

Note that a negative R^2 score indicates the predictive model is worse than a model predicting the average score for all values. With 10 of 12 models having negative R^2 scores and the other two having lower R^2 scores than any reasonably predictive model should, no model provides a reasonable prediction on ChatGPT-assigned grades. While it should be no surprise that the linear regression without regularization performed poorly due to the ratio of features to samples, the lasso and ridge models could have addressed this issue without the need to perform feature selection or projection. Since this was not the case, we needed to systematically reduce the number of features in each model to potentially improve the regressions. A common informal guideline is to have at least 10 samples per feature (Ogundimu et al., 2016). We thus needed to systematically reduce the number of features in each model to no more than seventeen given our set of 171 ChatGPT-assigned grades.

4.3 Regression Models After Feature Selection

We utilized the SelectKBest function in Scikit-Learn for a number of features between two and seventeen for each set of linguistic features. All average cross-validation R^2 scores were negative for all sets of linguistic features as shown in Table 2. No subset of selected features produced any reasonably predictive models.

Table 2

Range of average R^2 score from a five-fold cross-validation (Avg X-Val) for each linear regression built by applying SelectKBest on each of the four subsets of linguistic features.

Model	# of	TAACO	TAALED	TAALES	TAASC
Type	Features	Avg X-Val	Avg X-Val	Avg X-Val	Avg X-Val
linear	2-17	[-0.505, -0.050]	[-0.335, -0.137]	[-0.511, -0.052]	[-0.347, -0.112]
ridge	2-17	[-0.317, -0.042]	[-0.29, -0.127]	[-0.308, -0.052]	[-0.196, -0.066]
lasso	2-17	[-0.076, -0.076]	[-0.076, -0.076]	[-0.076, -0.076]	[-0.076, -0.076]

4.4 Regression Models After Feature Projection

We then utilized Principle Component Analysis to create linear regression models for 2 to 17 combined features on each subset of linguistic features. This again proved fruitless with all average cross-validation R^2 scores reporting negative values as shown in Table 3.

Table 3

Range of average R^2 score from a five-fold cross-validation (Avg X-Val) for each linear regression built by applying Principal Component Analysis on each of the four subsets of linguistic features.

Model	# of	TAACO	TAALED	TAALES	TAASC
Type	Features	Avg X-Val	Avg X-Val	Avg X-Val	Avg X-Val
linear	2-17	[-0.229, -0.102]	[-0.407, -0.133]	[-0.493, -0.074]	[-0.228, -0.055]
ridge	2-17	[-0.229, -0.102]	[-0.308, -0.133]	[-0.491, -0.074]	[-0.227, -0.055]
lasso	2-17	[-0.076, -0.076]	[-0.076, -0.076]	[-0.076, -0.076]	[-0.076, -0.076]

4.5 Classification Models

Since we were unable to create a predictive linear regression model using the linguistic features, we relaxed the target continuous grade to classes of grades: A as 90-100, B as 80-89, C as 70-79, D as 60-69, and F as 0-59. Distribution of grade classes appears in Figure 2. As a baseline estimator for accuracy, we note that a naive model guessing A for all letter grades would have about a 53% accuracy.

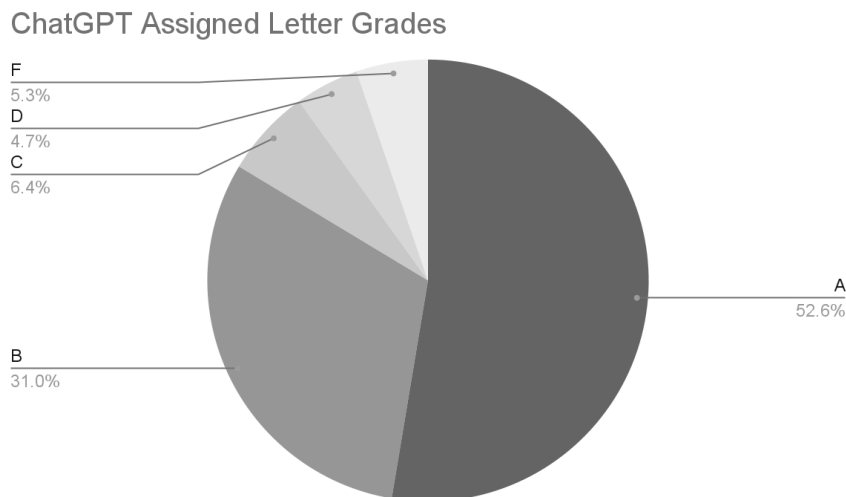


Figure 2: Distribution of ChatGPT-assigned letter grades for student discussion posts.

As Random Forest models can be sensitive to the data used for testing due to their tendency to overfit, we performed 1000 random forest models on each subset of linguistic features to determine both how stable and accurate the generated models would be. A grid search cross-validation (5-folds) was performed on each random forest model to improve each model's accuracy. Descriptive results of the best model accuracy scores appear in Table 4. While some models reached acceptable accuracy scores, the majority of models' accuracy hovered around the naive "all-A" model's 53% accuracy as seen by the 25% and 75% quartile thresholds for each subset of linguistic features. Moreover, the high standard deviation of 5-7% illustrates the instability of a particular model that could be used to accurately predict new data.

Table 4

Descriptive statistics for 1000 models' accuracy scores for each of the four subsets of linguistic features.

	TAACO	TAALED	TAALES	TAASC	All Features
mean	0.454	0.458	0.530	0.519	0.477
stdev	0.0506	0.0595	0.0726	0.0683	0.0669
min	0.265	0.265	0.206	0.324	0.206
25%	0.412	0.412	0.471	0.471	0.441
50%	0.441	0.471	0.529	0.529	0.471
75%	0.500	0.500	0.588	0.559	0.529
max	0.618	0.647	0.794	0.735	0.706

4.6 Summary of Results

Despite systematic attempts to improve the predictive power of a linear regression to predict ChatGPT-assigned grades on our set of data, no acceptable model emerged. Even when relaxing the predictive target to categorical letter grades, no acceptable predictive model emerged. This suggests that linguistic features, which can be used to reliably predict expert-assigned grades on writing, cannot alone predict ChatGPT-assigned grades. Specific results associated with each research question are provided below.

RQ1: Is there evidence that ChatGPT 3.5/4o assigns random grades to student writing?

All K-S hypothesis tests rejected the null hypothesis that the assigned grades were from the specific negatively-skewed distribution with p-values less than 1E-10. This is strong evidence that the ChatGPT-assigned grades were not randomly assigned from a skewed distribution and that some pattern may exist to predict how ChatGPT assigned grades.

RQ2: Can a model using a large set of linguistic features traditionally used in Automated Essay Scoring be used to predict scores that ChatGPT assigns to student writing?

Our data does not support the claim that a linear regression model using all linguistic features of a particular type can be used to predict scores that ChatGPT assigns. With all models having negative R^2 scores, no linear regression model applied to our data provided a reasonable prediction on ChatGPT-assigned grades.

RQ3: Can a model using a select subset of linguistic features traditionally used in Automated Essay Scoring be used to predict scores that ChatGPT 3.5 assigns to student writing?

Our data does not support the claim that a linear regression model using a selection of linguistic features of a particular type can be used to predict scores that ChatGPT assigns. All average cross-validation R^2 scores were negative for all selected sets of linguistic features, and thus, no linear regression model applied to our data provided a reasonable prediction on ChatGPT-assigned grades.

RQ4: Can a model using a linear combination of linguistic features traditionally used in Automated Essay Scoring be used to predict scores that ChatGPT 3.5 assigns to student writing?

Our data does not support the claim that a linear regression model using a projection of linguistic features of a particular type can be used to predict scores that ChatGPT assigns. All average cross-validation R^2 scores were negative for projections of sizes two to eight for each collection of linguistic features, indicating that no linear regression model applied to our data provided a reasonable prediction of ChatGPT-assigned grades.

RQ5: Can a classification model using linguistic features traditionally used in Automated Essay Scoring be used to predict letter grades that ChatGPT 3.5 assigns to student writing?

Our data does not support the claim that a classification model using linguistic features can be used to predict letter grades that ChatGPT assigns. The majority of constructed models' accuracy hovered around the naive "all-A" model's 53% accuracy and the high standard deviation of 5-7% illustrates the instability of a particular model that could be used to accurately predict new data.

5. Discussion

The present study aimed to determine whether the same linguistic characteristics that have historically predicted essay scoring could account for the grading behavior of ChatGPT (3.5/4o). Despite extensive modeling using indices that included lexical sophistication, syntactic complexity, and multiple regression models, no linguistic predictors consistently explained the variance in the AI-generated grades across any of the included disciplines or coursework levels. These findings indicate that while ChatGPT's grading is not random, its decision processes differ fundamentally from the feature-based results of previous AES systems.

Previous AES frameworks have demonstrated strong relationships between their identified linguistic variables and human-assigned scores. Early AES systems like Project Essay Grade and IntelliMetric demonstrated correlation using essay length, syntax, vocabulary, and discourse indicators to establish an assumption that linguistic features serve as measurable proxies for writing quality (Atali & Burnstein, 2006; Burstein, 2003; Rudner et al., 2006). Feature-based scoring later became more salient, showing AES models integrating lexical diversity and syntactic complexity (Kyle & Crossley, 2014; Kyle, 2018; Vajjala, 2018).

Alternatively, the negative and near-zero R^2 values observed in this study across every feature set suggest that the generative mechanisms of ChatGPT 3.5/4o did not align with these

established linguistic cues in its evaluation process. Instead, internal processes developed in the pre- and post-training production of the LLM underlying ChatGPT produce results that are opaque (with respect to AES and AWE tasks) despite our efforts to uncover them. This brings into question the validity, fairness, and transparency that have been a focus of previous AES development. This points to previous studies which have indicated that AI grading, though perceived by students as accurate, offers shallow feedback focused on surface correctness rather than deep understanding, underscoring a misalignment with instructional values (Almegren et al., 2024; Kaliisa et al., 2025; Velez et al., 2025).

Our finding that standard lexical, syntactic, and cohesion indices explain little of ChatGPT's scores also complicates claims of construct validity if instructors adopt LLM scoring. If an automated grader's outputs are not systematically linked to interpretable linguistic constructs known to human raters, it becomes difficult to justify the tool's use for high-stakes or formative assessment without additional transparency or validation. Writing-analytics scholars have urged that such tools be evaluated not only for accuracy but for fairness, ethical harms, and alignment with instructional constructs — concerns echoed in the *Journal of Writing Analytics* taxonomy and critiques of uncritical analytics adoption (Lang et al., 2019; Palmquist, 2019).

Taken together, the numerous models and null results may signal the need for a shift in how we understand and utilize AI-driven grading. Earlier AES systems were constrained by linguistic models, and generative AI systems, such as ChatGPT, appear to make judgments that are consistent but theoretically unknown. This demonstrates the need for further exploration and cautious integration of GenAI tools in assessment as they show the capacity to mimic consistency in scoring to appear meaningful but do not appear to be tied to an explicit theory of writing quality.

6. Conclusion

This paper reported an exploratory study in which we investigated the extent to which linguistic features predict scores from ChatGPT on university students' written assignments. Our findings indicate that while ChatGPT 3.5's grading of student writing is not random, it also does not follow the linguistic features that have traditionally underpinned AES systems as discussed in the literature review. Despite applying regression models across multiple sets of lexical, syntactic, and cohesion indices drawn from established tools (TAALES, TAASSC, TAACO, and TAALED), no reproducible patterns emerged. These results suggest that ChatGPT evaluated student writing according to mechanisms that remain opaque. This is not to say, however, that ChatGPT 3.5/4o could not be trained to score essays along linguistic features, but rather that it does not appear to do so without instructions specifying that it should. Therefore, while it is possible that generative AI has the potential to be used for automatic essay grading, we argue that it would need further training before it grades in ways similar to prior automatic grading.

While these findings provide an interesting starting point for this area of discussion, the study is not without limitations. Given the number of samples we had for ChatGPT 3.5 and 4o responses, we could not create models with more than about 10 features without risking overfitting. It is possible that more sophisticated models from sets of linguistic features may be able to predict how ChatGPT 3.5/4o assigned grades. We also only constructed linear regressions and acknowledge that alternative regressions may provide stronger predictive power on how ChatGPT assigned grades. It is also possible that other models of generative AI (and even newer models of ChatGPT) may be predicted using the methods and features we analyzed. We also acknowledge that no attempt was made to determine the stability of ChatGPT-assigned grades on

the same student response. Additionally, because courses were drawn from the project members' teaching loads, the dataset constituted a convenience sample. While this limits claims of generalizability, the varied disciplines and levels included also ensured a broad range of student writing contexts.

Similarly, our regression models treated essays as independent observations, but writing assessment research demonstrates that essay data are typically hierarchical—nested within raters, courses, tasks, or demographic groups. Such nested structures can distort effect sizes and produce misleading inferences when analyzed with single-level models. As McCallum (2019) noted, mixed-effects and multifactorial modeling approaches provide a more appropriate statistical framework for capturing these dependencies and for linking linguistic features to functional or rhetorical constructs. Future work should therefore reanalyze LLM scoring with mixed-effects models to determine whether any systematic linguistic predictors emerge once rater and task variance are properly accounted for.

Despite these limitations, we believe that the study provides a valuable contribution to conversations in writing research and assessment. We showed that ChatGPT does not simply replicate the feature-based logics of AES or the evaluative patterns of human raters. Rather, it appears to assign scores through processes that remain obscured, even as it is increasingly pitched to instructors as a way to save time with grading. By demonstrating this divergence, we establish a baseline that can help writing researchers, educators, and institutions more critically evaluate the claims and risks associated with GenAI systems in assessment contexts.

Future research should extend this work by exploring non-linguistic or rhetorical predictors of ChatGPT grading, such as citation use or alignment with assignment prompts (e.g., text similarity measures between student output and assignment prompt), and by testing stability across repeated scorings. Comparative studies of newer or fine-tuned models will also be important to assess whether generative AI can be trained to score in ways more consistent with human or AES practices. Finally, as institutions consider the role of GenAI in writing assessment, further research must grapple with the pedagogical and policy implications of systems that provide evaluations without transparent criteria.

Author Note:

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. This study was reviewed and considered exempt from regulatory oversight by the university's institutional review board. See [https://GitHub.com/\[ANONYMIZED\]](https://GitHub.com/[ANONYMIZED]) for Python code. Additional anonymized data files are available from the corresponding author upon reasonable request. The project was supported by the [blinded internal grant]. This work could not have been done without additional project team members [blinded].

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